Mortality Patterns in Romania 2000-2015: Infant Mortality, Adult Mortality and Causes of Death

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Abstract: This study documents the recent mortality patterns of Romanian population between 2000 and 2015. Despite the increasing life expectancy at birth, Romania is still witnessing one of the highest infant death rates and tuberculosis death rates in the EU. Based on the internationally available World Health Organization mortality database, this study focuses on three aspects of mortality data: infant mortality, adult mortality, and causes of death, using age-specific decompositions and multiple-decrement life tables. Results show disparities among age groups by sex in a fifteen-year period. The infant mortality and tuberculosis mortality have decreased significantly, yet tuberculosis remains a major death cause among young population. Mortality rates attributable to behavioural factors have been increasing. Moreover, there is an unusually high mortality rate among the 1986-1990 cohorts. These findings shed light on the health conditions in Romania before and after its accession to the European Union.

Keywords: mortality patterns, infant mortality, adult mortality, tuberculosis, ischemic heart disease, lung cancer, health disparities by sex

1. Introduction

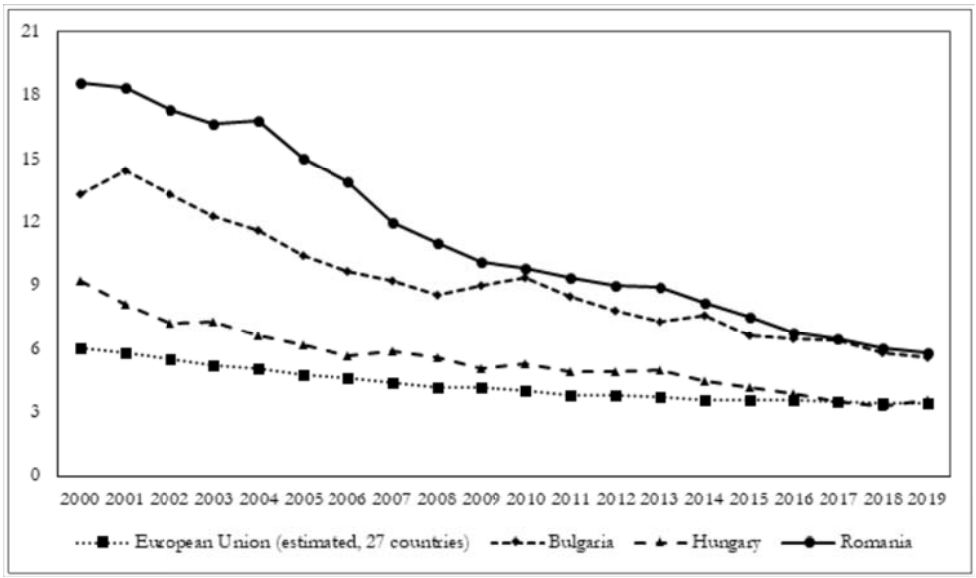
Romania has been known for its distinctive political and socio-economic characteristics from communism to the ongoing democratic transformation process since the 1990s. After joining the European Union (EU) in 2007, its demographic change continues to be a unique one. Its mortality data is particularly alarming within the EU level. As mortality is directly related to the socioeconomic conditions of a country, this study gives specific attention to the changes in mortality patterns before and after Romania's accession to the EU.

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In recent years, demographic studies on the Romanian or the broader Eastern Europe population mainly focus on the aging phenomena (Bodogai and Cutler 2014; Brie 2019; Coleman and Rowthorn 2011; Kluge, Goldstein, and Vogt 2019), declining fertility rates (Ciritel, De Rose, and Arezzo 2019; Jemna and David 2018), migration dynamics (Földes 2019; Horváth and Anghel 2009; Toşa et al. 2019) and life expectancy at birth (Dolea, Nolte, and McKee 2002; Jasilionis et al. 2011; Muntele et al. 2020). Studies that focus on mortality patterns are mainly from a comparative perspective (Duthé et al. 2017; Meslé 2004), with the exception of Dolea, Nolte, and McKee's study (2002) and Ursache and Muntele's study (2019), which specifically analyse the mortality patterns of Romania. Dolea, Nolte, and McKee (2002) looked into the mortality patterns of Romania as a means of describing life expectancy changes in the 1990s. By analysing major causes of death in Romania in that period, they found that the evolution in life expectancy in Romania after the transition was attributable to an interplay of economic and political factors as well as lifestyles related factors. Another case study in Romanian mortality was conducted by Ursache and Muntele (2019) examined the dynamics of infant mortality in Iaşi County in Romania using factorial analysis. In addition, the State of Health in the EU's Country Health Profiles by World Health Organization published in 2019 offered information on the health inequalities in Romania from different perspectives. All of these findings are valuable to the following discussions of this study.

Like other Central and Eastern European countries, Romania witnessed a slight increase of mortality rates around the 1990s due to medical crisis and the changing definition of live births (Ursache and Muntele 2019). This situation ameliorated somewhat after the turmoil period, with life expectancy at birth witnessing a gradual increase since 2000. However, Romania still experiences one of the highest infant mortality rates in the EU. The infant mortality rate is 18.6 deaths per 1,000 people in Romania, over three times higher than the average EU level in 2000, while it has decreased significantly to 5.8 deaths per 1,000 people in 2019. As shown in Figure 1, the infant mortality rate in Romania was strikingly high at the beginning but has been nearing the average EU level and the level of its neighbouring country Bulgaria in recent years. How does this change occur, and what factors contributed to the amelioration of infant mortality trend? It is thus necessary to look at the causes of death data for infant mortality to understand its patterns.

Figure 1. Infant mortality rates per 1,000 people, 2000-2019

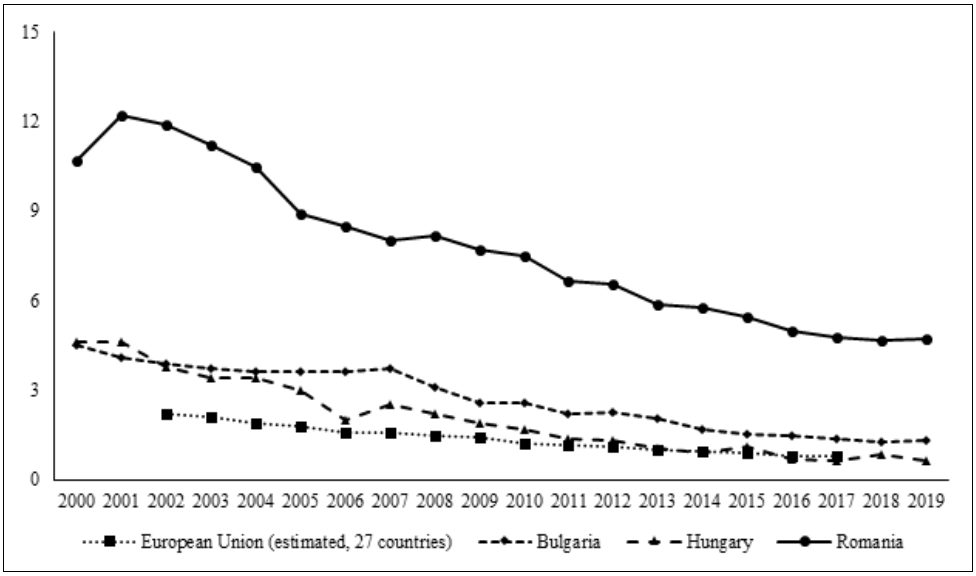


Source: Eurostat

On the other hand, the adult mortality rates of ischemic heart disease and stroke are also well above the EU level: ischemic heart disease mortality rate is almost three times higher in Romania, 550 deaths per 100,000 population in 2016, than in the EU as a whole; while the stroke mortality rate, 256 deaths per 100,000 population in 2016, is more than three times the EU average (World Health Organization Country Health Profile 2019). Lung cancer remains the most frequent cause of cancer deaths and its mortality rate has increased by 14% since 2000 (World Health Organization 2019). Romania also has the highest rate of tuberculosis cases among all EU countries, as shown in Figure 2.

Although the tuberculosis death rate has been decreasing in Romania since 2001, it is still ranked the highest in the EU, almost triple the average EU level and double its neighbouring countries like Bulgaria and Hungary. The strikingly high tuberculosis death rate leads us to think further about the mortality patterns of Romanian population in general. How does the age-specific mortality change in recent years? Moreover, in what way does the infant mortality trend differ from the adult mortality one?

Figure 2. Standardized tuberculosis death rates per 100,000 people, 2000-2017



Source: Eurostat

With the internationally available World Health Organization (WHO) mortality data, this study aims to document and analyse the patterns of infant mortality, adult mortality and the major causes of deaths and how they differ by sex and age groups for Romanian population between 2000 and 2015. Through age-specific decompositions and life table analysis, this study contributes to the Romanian literature with mortality information in an international database and offers a lens of the socio-economic development and health conditions in Romania.

2. Data and Methods

This study obtained the Romanian mid-year population, death counts, live births and causes of death data for the year 2000, 2005, 2010 and 2015 from the World Health Organization mortality database. These data cover information on age group, sex and classification of deaths identified by the 10th revision of the International Statistical Classification of Diseases (ICD-10). This study only focuses on the top three causes of diseases for infant mortality and adult mortality respectively, along with the tuberculosis deaths. The classification of causes of infant deaths ranks perinatal mortality (P00-96), pneumonia (J18) and congenital malformations of the circulatory system (Q20-

28) as the three highest categories. The perinatal mortality category entails certain conditions originating in the perinatal period. The category of pneumonia in this study does not include pneumonia due to other causes or in diseases classified elsewhere. The top three causes of adult mortality and tuberculosis are: ischemic heart disease (I20-25), stroke (I64), lung cancer (C34), tuberculosis (A15-19). As the mortality database classifies the causes of death data in detail, the related data were compiled based on the ICD-10 criteria before further analysis. All these data are openly available at https://www.who.int/data/data-collection-tools/who-mortality-database.

The main demographic methods employed in this study are age-specific rates and multiple decrement life table (MDLT). Age-specific rates were applied to analyse the age-specific mortality rates of Romanian population by sex between 2000 and 2015 and their changes in a five-year period. MDLTs were applied to analyse cause-eliminated life expectancy by sex for the year 2000, 2005, 2010 and 2015.

3. Results

3.1. Age-specific mortality changes by sex in Romania

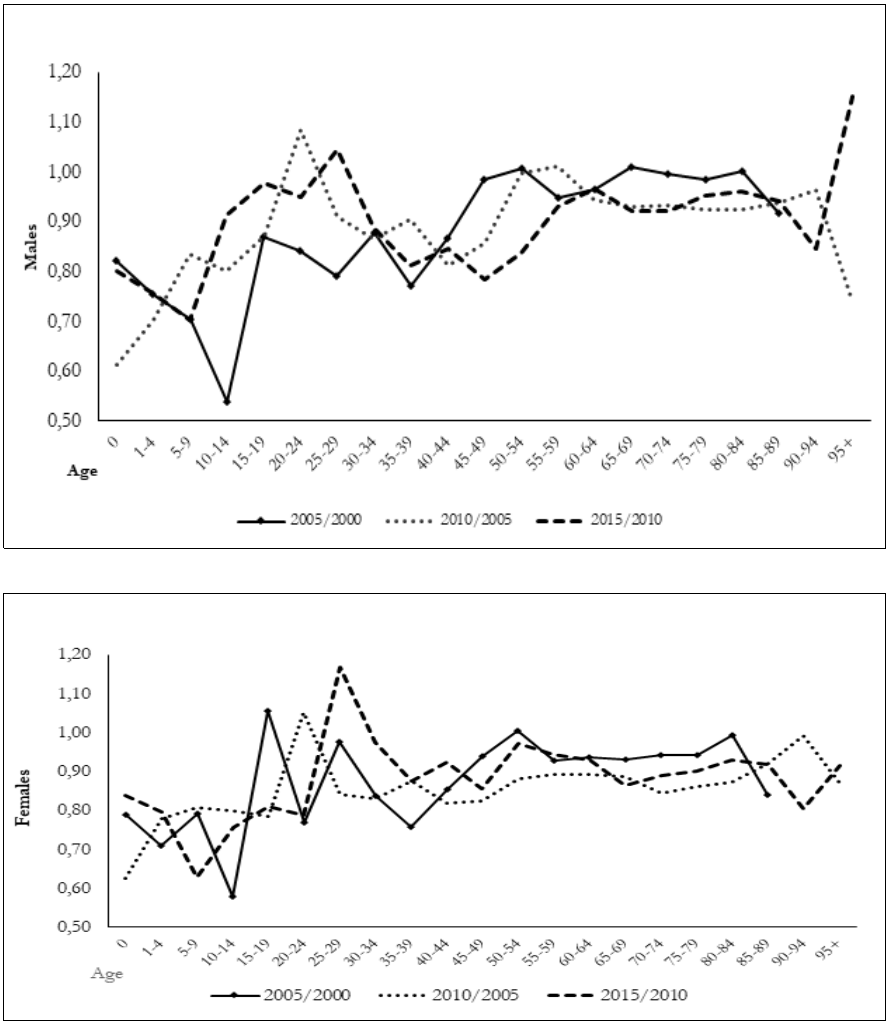
This study began with describing the mortality changes in Romania over time from 2000 to 2015. After estimating the age-specific mortality rates for the year 2000, 2005, 2010 and 2015, the ratio of death rates in each age group to the previous five-year period was calculated.

Figure 3 shows the ratio of age-specific death rates to the previous period with a five-year interval, that is, in 2005 to 2000, in 2010 to 2005, and in 2015 to 2010, for the Romanian male population and female population respectively. The trends in age-specific mortality share similarities between males and females. Overall, the age-specific rates have been decreasing for most ages except for a few deviations. The age-specific mortality rates for age 10-14 in 2005, for 35-39 in 2005, infant mortality in 2010 and for age 5-9 in 2015 have dropped significantly compared to previous years.

Nevertheless, it is worth noting that the 1986-1990 cohorts especially the female population has witnessed the increase of age-specific mortality rate at least since 2005, as shown by the ratio that exceeds 1.0 among age 15-19 in 2005, age 20-24 in 2010 and age 25-29 in 2015. Potential reasons for this phenomenon will be further discussed in the analysis of causes of death for adult mortality (see Figure 5). In addition, the age-specific mortality rates of younger ages before age 30 are more fluctuated; while the changing trend has been steadier among the elderly after age 55, except for the age 95 and over group of male population in 2010 and 2015, which shows a delayed mortality

effect for this cohort from 2010 to 2015. The significant decrease of infant mortality in 2010 and age-specific mortality rate for age 5-9 in 2015 shows an improvement of health conditions from previous years as well; it is possible that the reduced mortality rate for the cohorts born after 2005 is related to Romania's EU membership since 2007.

Figure 3: Changes of age-specific mortality rates in Romania by sex compared to the previous five-year period, 2000-2005, 2005-2010, 2010-2015

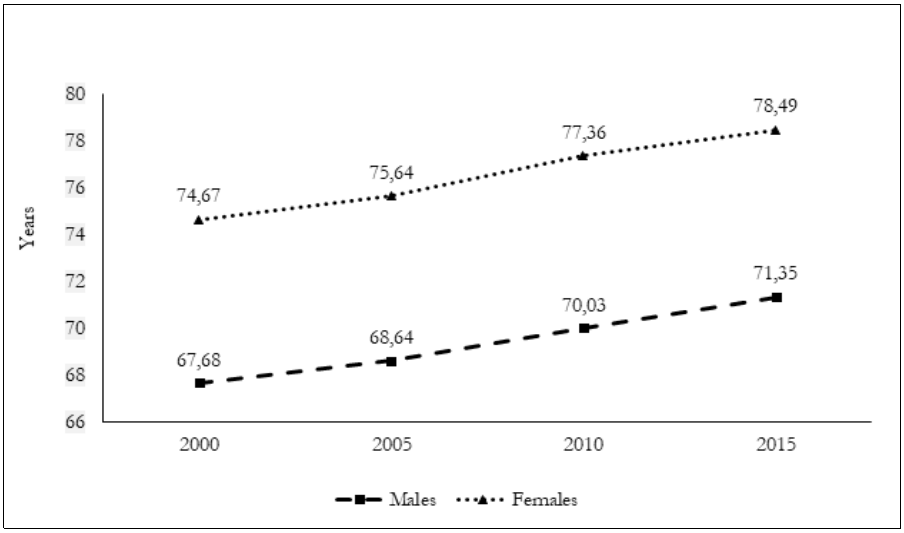


Source: WHO mortality database

3.2. Life expectancy at birth by sex in Romania

Figure 4 presents the results of life expectancy at birth by sex in Romania for the year 2000, 2005, 2010 and 2015 estimated by life tables.

Figure 4. Life expectancy at birth in Romania by sex, in the year 2000, 2005, 2010 and 2015



Source: WHO mortality database

The result shows that both the male and female population have improved their life expectancy at birth from 2000 to 2015, with an increase of approximately 4 years. The life expectancy at birth for both sexes in Romania has been increasing steadily at similar speed; however, it is still behind the average EU levels for both sexes, which is 77.7 years for males and 83.3 years for females in 2015 (Eurostat, 2020). Moreover, the life expectancy at birth for male population in Romania has a larger gap from the EU level (6.35 years less) than the female population (4.8 years less).[1]

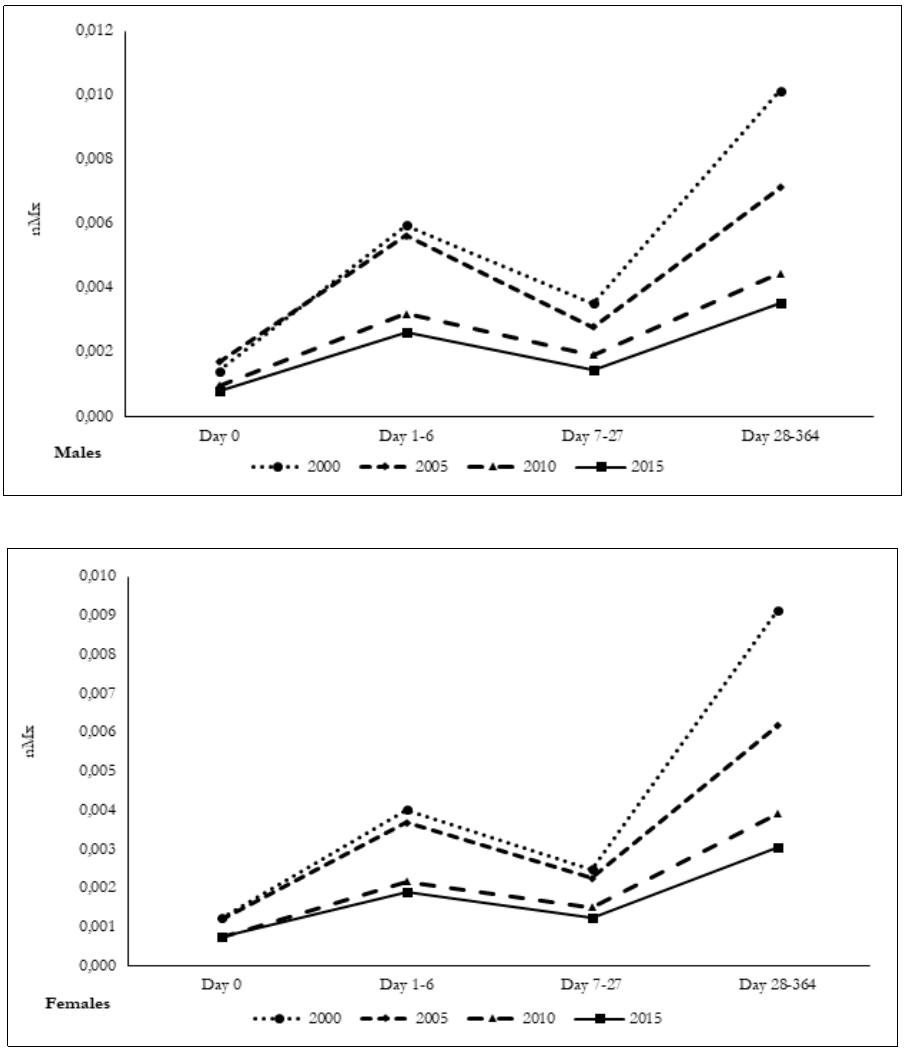
3.3. Infant mortality and its causes of death in Romania

In this analysis, I used the mortality data of infant deaths by sex during day 0, day 1-6, day 7-27 and day 28-364 for the year 2000, 2005, 2010 and 2015 from

[1] Data on the life expectancy at birth for the EU is drawn from the Eurostat database. Due to data completion, the average EU level data only includes 27 countries before Croatia joined the EU in 2013.

the WHO mortality database to plot the timeline of death occurrence (see Figure 5).

Figure 5. Timeline of infant mortality in Romania by sex, 2000-2015



Source: WHO mortality database

The mortality timelines for both sexes are in similar shape: most death cases occur during day 28-324. The death rates of day 0 and day 1-6 did not change much in a five-year period from 2005 to 2000 or from 2015 to 2010. The

infant mortality rate for males is higher than females and has a higher occurrence during day 1-6 in 2000 and 2005. On the other hand, the mortality rate of day 28-324 has also substantially decreased by years for both sexes. The gap of infant mortality between male infants and female infants has been reduced since 2010 especially for day 1-6. There is also a sign of significant improvement for the death rates of early infant mortality within a week in the 2005-2010 period.

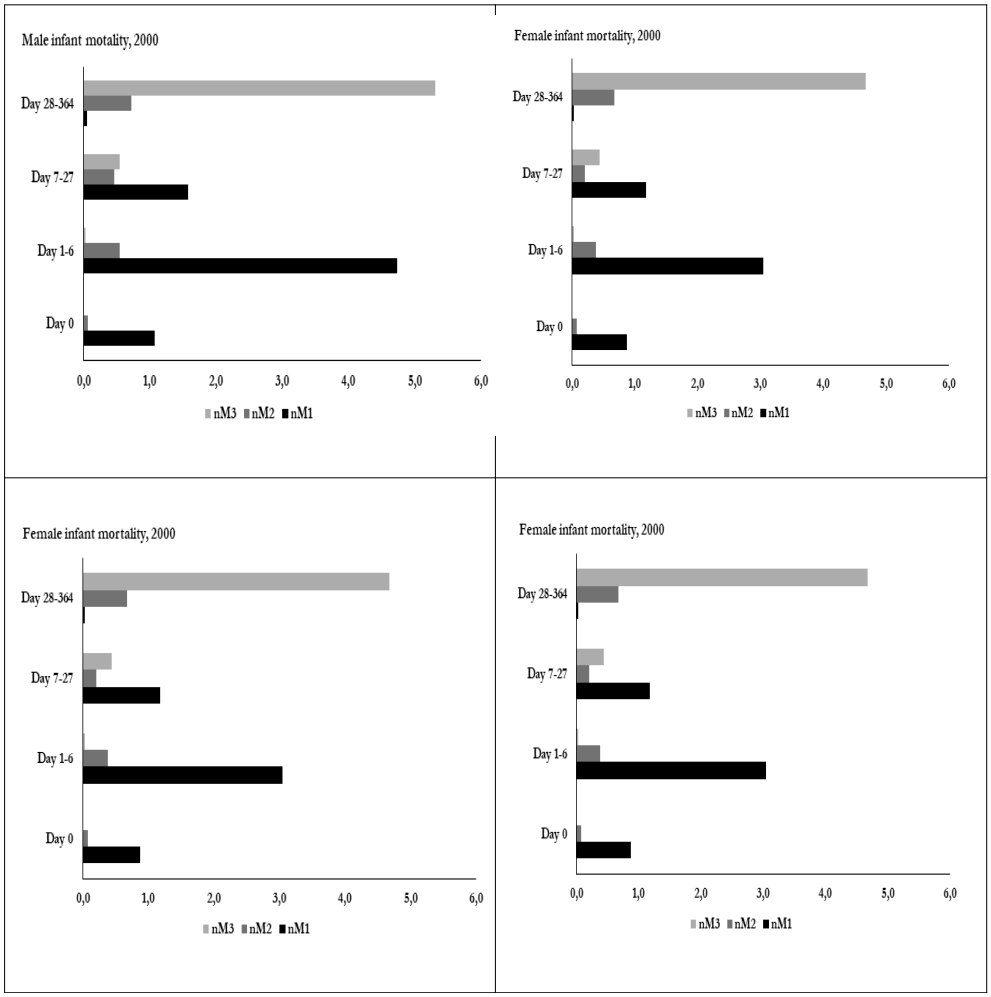
To capture the mortality patterns of infant mortality by sex, I analysed three major causes of death--perinatal mortality, congenital malformations of the circulatory system and pneumonia, and their occurrence time within 365 days in 2000 and 2015 (see Figure 6). $_nM_1$, $_nM_2$ and $_nM_3$ refer to the mortality rates attributed to perinatal mortality, congenital malformations of the circulatory system and pneumonia respectively.

As shown in Figure 6, perinatal mortality is the top cause of death for infants within 7 days while pneumonia is the top cause for infants of 28-364 days; congenital malformations of the circulatory system is the second major cause of death in day 1-6 and day 28-364. Male infants have higher mortality rate in all three causes than female infants, especially in perinatal mortality for day 1-6 in 2000 and 2015. According to the Eurostat data in 2015, the perinatal mortality rate for Romanian males which is 1.6 death per 1000 people, reveals a larger gap from the average EU level of 1.1 deaths per 1000 people, while the perinatal mortality rate for Romanian females is 1.1 deaths per 1000 people compared to the average EU level of 0.8 deaths per 1000 people (Eurostat, 2021). This discrepancy in perinatal mortality also helps explain the larger gap of life expectancy at birth from EU average level for Romanian males than females. On the other hand, the causes of death related to respiratory system in which group pneumonia belongs to[2], indicate a higher risk of death rates for Romanian male infants (1.0 deaths per 1000) than female infants (0.8 deaths per 1000) in 2015, which are well above the average EU level of 0.1 deaths per 1,000 people. The infant mortality rate attributable to respiratory system issue for both sexes is also much higher for Romania than Hungary and Bulgaria.[3]

[2] As Eurostat does not classify pneumonia (J18) for causes of death in infant mortality, the death rate data of the diseases of respiratory system (J00-99) is used for comparison here.

[3] In 2015, the death rate for infant mortality attributable to respiratory or circulatory system is 1.8 deaths per 1000 for Romania, 0.8 deaths per 1000 for Bulgaria, and 0.1 deaths per 1000 for Hungary (Eurostat, 2021).

Figure 6. Causes of death for infant mortality (per thousand) in Romania by sex, 2000 and 2015



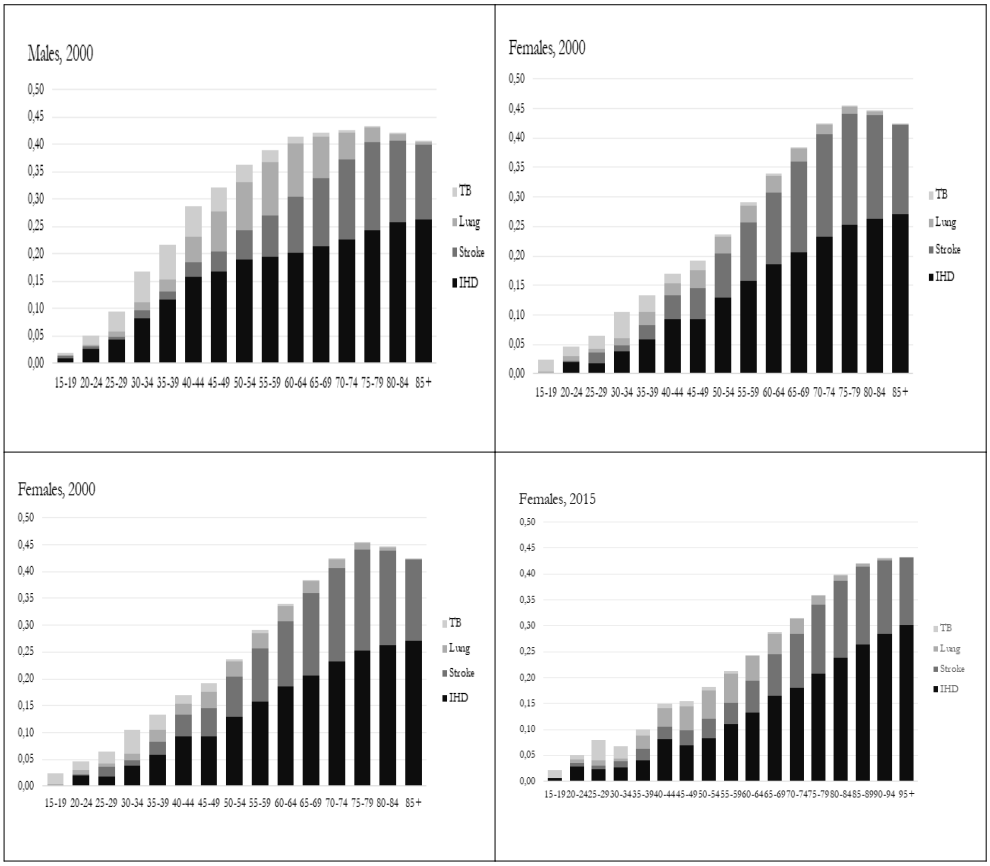
Source: WHO mortality database

Nevertheless, over a fifteen-year period the mortality rates of all three causes have diminished significantly, especially the perinatal mortality rate in day 1-6 and pneumonia mortality rate in day 28-364. The mortality rate of congenital malformations of the circulatory system did not decrease as much as the other two causes in this fifteen-year period. In other words, the decrease of perinatal mortality and pneumonia that contributed the most to the overall reduced infant mortality for both sexes in Romania from 2000 to 2015.

3.4. Adult mortality and its causes of death in Romania

For adult mortality, I analysed three major causes of deaths in Romania with the highest age-standardized mortality rates: ischemic heart disease, stroke and lung cancer, along with the top infectious disease--tuberculosis by age group and sex for the year 2000 and 2015 (see Figure 7). The results are displayed as the proportion that contributed to the adult mortality.

Figure 7. Proportion of deaths in adult mortality attribute to ischemic heart disease, stroke, lung cancer and tuberculosis, 2000 and 2015



Source: WHO mortality database

Note: "IHD" refers to ischemic heart disease, "Lung" is lung cancer and "TB" is tuberculosis. The age group 0-14 is left out and will not be discussed here as it does not fit the definition of adult mortality; age 60+ is included because of its close relation to the mortality rate of ischemic heart disease and stroke.

The results show that the mortality rates of ischemic heart disease and stroke increase by age in general. Lung cancer occurs the most from age 40 to age 75, while tuberculosis occurs in younger ages before age 60 and does not weigh much at elder ages. There is also a higher proportion of deaths attributed to lung cancer and tuberculosis among male adults than female adults in general, while stroke has a higher proportion in female adult mortality at higher ages especially after age 70. Although deaths from cerebrovascular diseases like stroke are common at elder ages, the high proportion of stroke and ischemic heart disease at younger ages is worthy of attention. Compared to other EU countries, Romania has the second highest cerebrovascular death cases in the EU only after Poland for both sexes of less than 65 years[4], and the second highest ischemic heart disease deaths for males and the third highest ischemic heart disease deaths for Romanian females[5] (Eurostat, 2021). Lung cancer deaths in Romania do not stand out by EU standards, but the higher occurrence among males has a significant effect on life expectancy at birth for males rather than for females, an issue which I will discuss later.

On the other hand, the trends of mortality rates in a fifteen-year period differ both by causes and by sex. The mortality rates attributed to ischemic heart disease and lung cancer have increased from 2000 to 2015 for both sexes, while the mortality rates attributable to stroke and tuberculosis have decreased for both sexes in the same period. In particular, the rising mortality rate among female adults is worth our attention. In 2000, lung cancer had a much lower effect on female adult mortality than male adult mortality, while in 2015 it affected female adults of age 40-74 more than before. The mortality rate attributed to lung cancer remains high for male adults, and has also increased for male adults of age 50-84 in 2015. Meanwhile, the increase of mortality rate of ischemic heart disease for both sexes is mainly for elder ages, while it is not as remarkable as the effect of lung cancer mortality rates in general. The stroke death rate remains relatively steady compared to lung cancer and tuberculosis.

For tuberculosis, the mortality rate has declined significantly from 2000 to 2015, especially for male adults in age 20-54. However, the tuberculosis death rate has increased for female adults of age 25-29. Tuberculosis is also a

[4] The cerebrovascular diseases (I60-I69) data is used for comparison because Eurostat does not specifically classify stroke (I64) as cause of deaths in the dataset. In 2015, the cerebrovascular deaths by age less than 65 years in Romania are 3046 deaths for males and 1514 deaths for females, while Poland the highest cerebrovascular deaths in the EU, 3280 and 1531 deaths for males and females respectively.

[5] In 2015, the ischemic heart diseases mortality by age less than 65 years in Romania are 6118 deaths for males and 1868 deaths for females, while Germany has the highest death cases in the EU for males and females, 10237 deaths and 2460 deaths respectively.

dominant cause of death for female adults of age 15-19 in 2000 and 2015, but it does not appear to have much effect on male adults in the same period. A further comparison with the tuberculosis death cases within the EU in 2015 indicates that female death cases (12 deaths) account for nearly half of total tuberculosis deaths in the EU (25 deaths) for the 25-29 age group, while the male deaths in Romania (20 deaths) make up more than half of total tuberculosis deaths (34 deaths) in the EU (Eurostat, 2021). On the other hand, there is no male death case while there are 2 female death cases for the 15-19 age group in Romania, compared to the total 2 male deaths and 3 female deaths in the EU. Although the proportion of tuberculosis deaths is not salient among the top causes of death in Romania, it is still very striking by EU standards.

As mentioned in the results of age-specific mortality rates, the 1986-1990 female cohorts have experienced a high mortality rate since 2005. The high mortality rate of tuberculosis for female adults in age 15-19 may be an explanation for this phenomenon. The unproportionally high tuberculosis mortality rate for females of age 25-29 in 2015 also corresponds to the high mortality rate of 1986-1990 female cohorts.

To analyse their effects on life expectancy at birth in Romania from 2000 to 2015, I applied the multiple-decrement life table (MDLT) technique for further analysis. The cause-eliminated life expectancy at birth by sex is shown in Table 1.

Table 1. Cause-eliminated life expectancy at birth by sex in Romania, 2000-2015

	Male				Female		
	2000	2005	2010	2015	2000	2005	2010
All causes	67.68	68.64	70.03	71.35	74.67	75.64	77.36
Ischemic heart disease	70.55	71.29	72.63	73.83	77.27	78.00	79.72
Stroke	69.01	69.89	71.19	72.36	76.33	77.24	78.78
Lung cancer	68.53	69.52	70.96	72.33	75.02	76.04	77.80
Tuberculosis	68.08	68.99	70.36	71.63	74.92	75.91	77.63

Source: WHO mortality database, 2019.

The results of cause-eliminated life expectancy at birth for the four causes align with the proportion of their impacts on the mortality rate of Romanian population--the top cause of death is ischemic heart disease, followed by

stroke and lung cancer; the effect of lung cancer is greater for males than females (World Health Organization, 2019). It is also worth noting that even though the tuberculosis death rate is the highest among the EU countries, it shows little effect on the mortality rate for both sexes when looking at the overall Romanian population. That said, the high risk of tuberculosis among the youths and age group of 25-29 deserves more attention from the public health sectors as the tuberculosis death rate in Romania is unusually high by the EU standards.

4. Discussion and Conclusion

This study uses the internationally available World Health Organization mortality database to document the age-specific mortality changes and key causes of death for infant mortality and adult mortality respectively in Romanian population between 2000 and 2015. In alignment with the Romanian report from World Health Organization Health Profile (2019), the results of life table analysis show that from 2000 to 2015, life expectancy at birth in Romania has increased by approximately four years. However, the changes of age-specific death rates for each five-year period indicate disparities by sex and age groups. To further investigate the primary causes of death for different age groups, infant mortality and adult mortality are examined respectively with each of their top three causes using age-specific decompositions. Additionally, infant mortality is decomposed by a timeline of day 0, day 1-6, day 7-27 and day 28-364, and multiple-decrement life table technique is applied for the causes of adult mortality.

The infant mortality data by year from the Eurostat database suggests that infant mortality rate has dropped significantly in the 2010s to a level closer to Bulgaria, though it is still above the average EU level. Yet the results of changes in age-specific rates by sex indicate that there is an unusual increase of mortality rate for the young population in Romania since 2005. The subsequent analysis of the causes of death in adult mortality reveals that the high mortality rate of age 25-29 in 2015, namely the 1986-1990 cohort, may be attributed to tuberculosis deaths among the young population. Moreover, the changes of age-specific mortality rates are fluctuated among the younger population before age 30, and the life expectancy at birth for Romanian males indicates a larger gap from the EU level than that of Romanian females does.

A closer look at the infant mortality and adult mortality brings up results of divergent mortality patterns. Overall, there is a substantial decrease of infant mortality from 2000 to 2015. The majority of infant deaths occurs during day 1-6 and day 28-364, in which perinatal mortality and pneumonia are

the main causes of death respectively. Therefore, the remarkable decline of perinatal mortality and pneumonia mortality rate contributed the most to the overall decrease of mortality in this period. In regard to adult mortality, there is a significant decline of tuberculosis deaths among the younger population especially for males from 2000 to 2015. Ischaemic heart disease, stroke and lung cancer occupy the top three causes of death. As ischaemic heart disease and stroke occurs more frequently among the elder population, lung cancer mortality is on the rise especially for the middle-aged population. There is also a higher proportion of deaths resulting from lung cancer among male adults than female adults, although the lung cancer mortality rate is also increasing among female adults. In fact, all of these causes discussed belong to preventable mortality and are attributable to behavioural factors such as obesity, alcohol and tobacco consumptions (World Health Organization 2019). In recent years, alcohol consumption and smoking have become a major public health issue not merely in Romania but also in the EU. However, as health spending in Romania remains the lowest in the EU (World Health Organization 2019), the underfunded condition of the health system may inhibit its ability to reduce preventable causes of death in the future.

Overall, these findings indicate a steady growth of life expectancy at birth in Romania overtime between 2000 and 2015. The perinatal mortality rate, pneumonia mortality rate among infants and tuberculosis mortality rate have decreased in general. There was also a reduction in stroke mortality with gender differences by age, as female adult mortality has a higher proportion in elder ages. These improvements are attributable to Romania's better access to outpatient healthcare after its accession into the EU in 2007 (Ioacara et al. 2019). Nevertheless, tuberculosis is still a major death cause among young population and remains a public health issue for Romania by EU standards. One of the reasons is that even though Romania has made progress in detecting and treating drug-susceptible tuberculosis (DS-TB), the accelerated migration of Romanian citizens across countries led to an increase of multidrug-resistant tuberculosis (MDR-TB) cases (Golli et al. 2019). On the other hand, mortality rates from ischemic heart disease and lung cancer have been increasing in recent years. Other cancer types not discussed here such as colorectal and breast cancers are also on the rise (World Health Organization 2019).

There are certain limitations of this study. First, this study only accessed internationally available data as the World Health Organization mortality database and English-written sources for references. It inevitably omits some details that only restricted data can offer and studies on this

subject that are not written in English. Second, the focus of this study is a descriptive analysis of the mortality patterns that can be inferred from the World Health Organization data. It does not intend to draw causal inference from the data on the relationship between mortality and socioeconomic conditions in Romania. However, the descriptive findings of this study may shed light on the health conditions in Romania before and after its accession to European Union. The findings also offer an overview of the trends and disparities in mortality patterns influenced by socioeconomic development and policies.

Moving forward, the infectious diseases among youths and the high proportion of preventable mortality attributable to behavioural factors may become key challenges to the public health system in Romania. Policy makers may need to implement effective measures to the preventions and survival from infectious diseases at birth and young ages. The increase of health expenditures per capita can help alleviate the incidence of deaths caused by preventable mortality (Simionescu et al. 2019). The socioeconomic factors underlying lifestyles, such as educational level, unemployment rates, gender, race and ethnic issues also need to be addressed to reduce the health disparities in the population.

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